POLI10302 / 20 CREDITS

Making Sense of Politics.

Course handbook

A guide to the questions, skills and activities that shape the module.

WORKING DRAFT

Teaching arrangements and assessment details will be confirmed on Canvas.



NAVIGATE THE HANDBOOK

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Welcome

Political questions deserve answers grounded in evidence.

Making Sense of Politics is an introduction to using data to explore political questions. You will learn to work with survey data, communicate a finding to a general audience and use an experiment to investigate a causal question.

This handbook introduces the module, its assessments and its weekly journey. Use it alongside Canvas, where the teaching team will share the current timetable, readings, assessment briefs and announcements.

The essentials

POLI10302 is a 20-credit module. Lectures introduce ideas and political questions; workshops give you space to practise with R and ask for help. No prior experience of programming or statistics is assumed.

Check your University timetable for times and locations, and Canvas for contacts, office hours, deadlines and the latest course information.

THE MODULE

Course description

How can we tell what people think, why politics changes and whether a political intervention makes a difference?

Politics is full of claims about what people believe and what causes them to act. Some are persuasive; others rest on anecdotes, selective examples or assumptions. This module gives you tools to ask sharper questions and examine those claims using evidence.

In the first part, we explore public opinion and learn how to find, prepare and describe political data. You will use R and create a clear visualisation, then explain a finding in a data journalism article for a non-specialist audience.

In the second part, we turn to experiments. You will learn why random assignment can help us understand cause and effect, compare outcomes across groups, and use a simple regression to estimate and interpret a treatment effect. You will present your analysis in a short research report.

A practical introduction

You do not need to arrive knowing how to code or feeling confident about statistics. Ideas are introduced step by step, and workshops give you time to try them for yourself. You will learn by asking questions, making mistakes, discussing interpretations and revisiting your work.

A habit of questioning

Data do not speak for themselves. Throughout the module, we ask where evidence comes from, what it measures, whom it represents and what it cannot tell us. The goal is to make arguments that are both informative and appropriately cautious.

WHAT YOU WILL LEARN

Course aims

Build the confidence to ask political questions and answer them with evidence.

Answer political questions

Turn curiosity into questions that can be investigated. Learn to choose relevant evidence, describe political patterns and distinguish a relationship from a convincing causal explanation.

Learn data science

Starting from scratch, find, prepare and analyse political data in R. Create visualisations with ggplot2, summarise patterns, compare groups and interpret a simple regression. You will also consider sampling, measurement, uncertainty and the limitations of your data.

Communicate with purpose

Make a clear and appropriately qualified argument for different audiences. Practise writing a data-driven article for the public and an experimental report for an academic reader, supported by relevant research.

Build skills for your future

Develop practical examples of working with data, evaluating claims and explaining findings. These skills can support further study and work in research, journalism, public policy and many other fields.

The formal module learning outcomes and University regulations are available through the University's published module information.

YOUR WORK

Assessments

Two assignments give you opportunities to use evidence for different audiences.

Assessment 1 / Data journalism article

1,000 words ± 10% · 35% of the module grade

Choose a political question and investigate it using a YouGov political tracker. Analyse the data in R and include at least one original figure made with ggplot2. Write for a general audience, explain your finding and its limitations, and use at least two academic journal articles or books to provide context.

Workshop 5 in teaching Week 6 is a supported session for developing your article and asking questions.

Assessment 2 / Experimental report

2,000 words ± 10% · 65% of the module grade

Write a research report in the style of an academic article, using an assessment dataset and a question provided by the teaching team. Develop a hypothesis from academic research, analyse the experiment in R and present at least one table or figure. Interpret the treatment effect and its uncertainty, and discuss what the results mean and their limitations. Use at least five academic journal articles or books.

Workshop 10 in teaching Week 11 is a supported session for developing your report.

Before you submit

The current draft briefs specify one PDF uploaded to Canvas, with your student ID, word count and reference list. Check the final briefs on Canvas for confirmed requirements, questions and deadlines; dates are not yet confirmed.

Assessment 1 brief · Assessment 2 brief

BEYOND THE MODULE

Employability

The work you do here gives you practical examples to draw on in applications and interviews.

Employers in research, government, journalism, charities and business use evidence to understand problems and make decisions. This module introduces skills that travel across those settings, while giving you a chance to explain what you have done and why.

Find and work with evidence

Locate data, assess where they came from and prepare them for analysis. Using R scripts helps you record the steps you took and revisit your work.

Think critically

Compare groups, identify limitations and distinguish correlation from causation. Learn to interpret uncertainty instead of treating every number as a definitive answer.

Explain what you find

Create a readable graphic, write for different audiences and describe a result in plain language. A clear explanation includes what the evidence does and does not support.

Use your assignments as examples

Assessment 1: survey data analysis in R; data visualisation with ggplot2; communication of quantitative findings to non-specialists.

Assessment 2: linear regression in R; causal inference from experiments; statistical reporting.

At the guest event, you can explore how professionals use quantitative skills in their work and ask about routes into different careers.

HOW TO MAKE THE MOST OF THE MODULE

Course expectations

You are here to learn. Regular practice, curiosity and asking for help matter more than arriving with experience.

Take part and practise

Attend lectures and your assigned workshops, engage with the questions and activities, and work through the practical exercises. Workshop materials are designed for you to work through at your own pace, with support from your tutor.

Return to the code between sessions. Try running it, changing one thing at a time and explaining what the output means. Save your scripts so you can revisit your steps.

Prepare and seek feedback

Check Canvas regularly for announcements, readings, assessment guidance and any changes. Before an assessment-support session or office hour, bring your question, relevant data or code, and a specific point you would like to discuss.

Work responsibly and support one another

Discuss ideas respectfully, listen to different perspectives and give others room to contribute. Acknowledge the sources you use, represent evidence honestly and follow the University's current guidance on academic integrity and permitted use of tools.

If you need help

Ask your workshop tutor or contact the teaching team through the details on Canvas. For submission problems, wellbeing, mitigating circumstances or other University support, contact Student Support. Check current University guidance for extensions and assessment rules.

Week 1

GETTING STARTED

Why use numbers to understand politics?

LECTURE

Welcome and introduction

We begin with political questions and the role of evidence in answering them. We consider what quantitative data can reveal, what they can miss and how the module's two parts fit together.

WORKSHOP

No workshop this week. Use this time to find the module on Canvas, check your timetable and familiarise yourself with the course materials.

By the end of the week

Identify a political claim you would like to investigate. What evidence would help you assess it?

Questions to think about

What makes a claim testable? Think about a political question, the evidence you would need and the limits of that evidence.

Before your next session: check Canvas for the current reading, materials and preparation.

Week 2

DATA SCIENCE FUNDAMENTALS

How do we know what people think?

LECTURE

Surveys, samples and political evidence

We explore how surveys turn individual responses into evidence about public opinion. We ask how samples are recruited, what it means for a sample to be representative and where to find political data.

WORKSHOP

Workshop 1: Getting started with data and R. Read a small table of YouGov results, make a simple calculation in R and save your first script.

By the end of the week

Describe what a survey result measures and whose views it is intended to represent.

Questions to think about

Who was asked, how were they selected and what exactly was the question? How might those choices affect an interpretation?

Before your next session: check Canvas for the current reading, materials and preparation.

Week 3

DATA SCIENCE FUNDAMENTALS

How do we know whether opinion is changing?

LECTURE

Measurement and trends

We consider how political attitudes are measured, how a concept becomes a variable and how comparisons over time can help us identify change. We also ask when changes in wording or measurement complicate a comparison.

WORKSHOP

Workshop 2: Prepare data. Begin finding and importing data, then inspect and prepare it in R. Practise selecting, filtering and cleaning data for analysis.

By the end of the week

Check that observations being compared measure the same thing in a meaningful way.

Questions to think about

Does a change in a number reflect a change in attitudes, or could a change in the question or sample also matter?

Before your next session: check Canvas for the current reading, materials and preparation.

Week 4

DATA SCIENCE FUNDAMENTALS

Who thinks what?

LECTURE

Groups and descriptive statistics

We use descriptive statistics to explore differences between groups. We consider proportions and other summaries, and ask how to describe a pattern without claiming more than the data support.

WORKSHOP

Workshop 3: Analyse data. Summarise political data in R, compare groups or periods and identify a finding worth explaining.

By the end of the week

Describe a difference in plain language, including the groups and units being compared.

Questions to think about

Which groups are being compared, and how large is the difference? What other information would help you understand it?

Before your next session: check Canvas for the current reading, materials and preparation.

Week 5

DATA SCIENCE FUNDAMENTALS

What makes a visualisation trustworthy?

LECTURE

Visualisation and data journalism

We examine how design choices shape the way readers understand evidence: labels, scales, colour and graphical integrity all matter. The second part introduces Assessment 1 and how to turn a finding into a clear article.

WORKSHOP

Workshop 4: Visualise data. Create, refine, label and export a chart using ggplot2.

By the end of the week

Choose a visual form that makes the comparison clear and gives readers the information they need.

Questions to think about

Can a reader identify the source, units and comparison without asking you? What would change if the axis or scale were different?

Before your next session: check Canvas for the current reading, materials and preparation.

Week 6

EXPERIMENTAL POLITICAL SCIENCE

How can an experiment tell us what causes what?

LECTURE

Experiments and causal questions

We introduce counterfactual thinking, treatment and control groups, and random assignment. We ask why association alone may not tell us what caused an outcome, and how an experiment can help.

WORKSHOP

Workshop 5: Assessment 1 support. Bring your question, data, code or a draft figure. Use the session to develop your article and troubleshoot with your tutor.

By the end of the week

Explain the difference between a descriptive pattern and a causal claim.

Questions to think about

What would the same person or group have experienced without the treatment? Why is that counterfactual difficult to observe?

Before your next session: check Canvas for the current reading, materials and preparation.

Week 7

EXPERIMENTAL POLITICAL SCIENCE

How do we measure an effect?

LECTURE

Comparing treatment and control

We define outcomes and treatment effects, compare group means and consider why an estimated effect can differ across samples. We also ask whether the size of a difference matters in political terms.

WORKSHOP

Workshop 6: Describe experimental data. Identify the treatment and outcome, calculate group means and compare the results.

By the end of the week

Interpret a difference between treatment and control groups in the units of the outcome.

Questions to think about

Which outcome is being compared? Is the estimated difference large enough to matter in the context of the question?

Before your next session: check Canvas for the current reading, materials and preparation.

Week 8

EXPERIMENTAL POLITICAL SCIENCE

Why use regression to analyse experiments?

LECTURE

From group differences to regression

We introduce simple linear regression as a way to estimate an experimental treatment effect. We interpret the intercept and treatment coefficient and connect the model to a difference in group means.

WORKSHOP

Workshop 7: Estimate treatment effects. Fit a simple model using lm() in R and interpret its coefficients.

By the end of the week

Explain what a treatment coefficient says about the difference between groups.

Questions to think about

What do the intercept and treatment coefficient represent? How does the coefficient relate to the group means?

Before your next session: check Canvas for the current reading, materials and preparation.

Week 9

EXPERIMENTAL POLITICAL SCIENCE

How certain are we about an effect?

LECTURE

Uncertainty and differences in effects

We interpret confidence intervals and p-values alongside the size and direction of an estimate. We also consider why an average effect may not describe every group, and why subgroup claims need careful evidence.

WORKSHOP

Workshop 8: Interpret statistical evidence. Extract estimates and measures of uncertainty and practise explaining them; an interaction exercise may be offered as an extension.

By the end of the week

Describe an estimate and its uncertainty without treating statistical significance as the whole story.

Questions to think about

What range of effects is compatible with the confidence interval? How should uncertainty affect the language used to describe the result?

Before your next session: check Canvas for the current reading, materials and preparation.

Week 10

LOOKING AHEAD

Where can these skills take you?

LECTURE

Guest event and professional practice

Hear from professionals about how they use quantitative and analytical skills. Reflect on the questions they ask, the evidence they use and the ways they communicate their findings.

WORKSHOP

Workshop 9: Communicate research findings. Prepare a results table or figure, document your analysis and practise writing up an experimental result.

By the end of the week

Connect a skill you have practised to an example you could discuss in an application or interview.

Questions to think about

How do professionals move from a broad question to a useful analysis? Which skills have you practised that could help them?

Before your next session: check Canvas for the current reading, materials and preparation.

Week 11

LOOKING AHEAD

How do I bring my analysis together?

LECTURE

No lecture this week

There is no lecture this week. Use the time to review the assessment brief and bring together your question, literature, analysis and interpretation.

WORKSHOP

Workshop 10: Assessment 2 support. Bring your data, script, results or draft and use the session to work on the experimental report with help from your tutor.

By the end of the week

Check that your report answers its question, interprets uncertainty and explains the study's limitations.

Questions to think about

Does the evidence answer your research question? Are the treatment, outcome, estimate and limitations clear to a reader?

Before submitting: check Canvas for the final brief, deadline and requirements.

KEEP EXPLORING

Further reading

Continue at your own pace. These resources are starting points, not a required reading list.

Learn more R

Wickham, Çetinkaya-Rundel and Grolemund, R for Data Science (2nd edition). A practical guide to importing, transforming and visualising data with R and the tidyverse. Read online.

Make better visualisations

Kieran Healy, Data Visualization: A Practical Introduction. A guide to thinking about and making effective graphics with R. Explore the book.

Think about causal questions

Nick Huntington-Klein, The Effect: An Introduction to Research Design and Causality. A next step in thinking about research design and causal inference, including approaches beyond experiments. Read online.

Explore experiments in depth

Alan Gerber and Donald Green, Field Experiments: Design, Analysis, and Interpretation. A fuller introduction to designing, analysing and interpreting experiments in the social sciences. Book information.

Find political data

Explore YouGov's political trackers and consider a question of your own. Before analysing any dataset, check how it was collected, what its variables mean and who is represented.

For books requiring access, check the University of Manchester Library.